What a Remaining-Useful-Life Interval Can Prove: Coverage, Width, and the Calibration Object

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***Abstract*—Prediction intervals for remaining useful life (RUL) are compared on a coverage rate close to nominal and a width as small as possible. This paper measures how much of that pair is information and how much is reporting convention, using 653 C-MAPSS test engines and 2 587 N-CMAPSS queries under one fixed protocol. Three findings. Scoring coverage on the integer grid instead of by randomised probability integral transform inflates it by $+0.011$ to $+0.044$ at the 90 % level and $+0.074$ to $+0.120$ at 50 %, independent of the model. At this sample size an RUL bin of 11 to 82 engines detects only deviations of 0.14 to 0.44, and 143 engines per bin are needed for 0.10, so most published comparisons of conditional coverage are unresolved by construction. And the object that is calibrated has to be the object that is deployed: a controlled experiment holding backbone, engines, centre, score scale and tail allocation fixed finds that matching them saves 2 to 11 % of width depending on the fleet, with the sign reversing on one fleet. The same control shows that a free tail allocation saves 16–22 % of width in a simulation with known truth but only 0–5 % on the benchmark, and that making its selection independent of the scores it is applied to costs a further 13 %. No conformal construction here is sharply better than a 66-parameter parametric layer; what separates the constructions is measurable, and what separates the reporting conventions is larger than the differences they are used to rank.**

***Index Terms*—Remaining useful life, uncertainty quantification, conformal prediction, calibration, coverage, measurement uncertainty, prognostics and health management.**

I. Introduction

An RUL interval is reported as a coverage and a width. The coverage is a frequency on a lattice-valued target capped by the benchmark; the width is a mean in cycles; and the interval is built from a residual sample whose composition depends on which model was calibrated. Each of these is a measurement decision, and this paper asks how much the reported numbers move when they are made differently on the same predictions.

The question matters because the differences that published tables rank are of the same size as the differences these conventions produce. On the four C-MAPSS fleets the conformal constructions compared here span 4.6 cycles of mean width and 0.03 of coverage; the integer-versus-PIT convention moves coverage by up to 0.044 on the same intervals, and the choice of calibrated object moves width by up to 4.2 cycles. A reader cannot tell from a reported pair whether a ranking reflects the method or the convention.

A. What this paper does and does not claim

It does not propose a new interval construction. The construction used here — split conformal prediction with signed scores and an asymmetric tail allocation — is the one of [12], [15], applied with the calibration sample matched to the deployed ensemble. Section III-A states its guarantee with attribution and states what the guarantee does not cover.

What the paper does is measure. Four results are established, each from a controlled comparison on fixed predictions:

1) **The coverage rule is worth up to 0.044.** Integer-endpoint scoring and randomised PIT are two ways of scoring the same predictive distribution; on the frozen layer they differ by $+0.011$ to $+0.044$ at 0.90 and $+0.074$ to $+0.120$ at 0.50, always in the same direction, with no dependence on the predictor (Table IV).
2) **The benchmark cannot resolve most of what it is used to rank.** With 11–82 engines per RUL bin, the smallest deviation an exact binomial test detects at family-wise 5 % and 80 % power is 0.14–0.44; 143 engines per bin are needed for 0.10 (Table V).
3) **The calibrated object has to be the deployed object, and matching it is worth 2–11 %, not more.** Fixing backbone, engines, centre, score scale and tail allocation and varying only the calibration sample gives 2.7–10.8 % on the four fleets, with one fleet reversing when the tail is held symmetric (Table II).
4) **A free tail is not free.** In a simulation with known truth it saves 16–22 % of width at equal coverage; on the benchmark it saves 0–5 % and reverses on one fleet; and making its allocation independent of the order statistics it is applied to, so the bound applies as stated, costs 13 % of width (Section V-A).

Section II treats the scoring rule and the detectability of a coverage claim, Section III the calibrated object, Section IV the protocol, and Section V the measurements.

II. What a Coverage Number Can Resolve

A. Two ways to score the same distribution

Let F_i be a predictive distribution over the integer grid $\{0, \dots, K\}$, $K = 125$, and let Y_i be the realised RUL. An interval $[L_i, U_i]$ covers when $L_i \le Y_i \le U_i$.

Integer endpoints. L_i and U_i are integers and coverage is the frequency of $L_i \le Y_i \le U_i$. This counts the whole atom at Y_i.

Randomised PIT. Draw $V_i \sim \mathrm{U}(F_i(Y_i - 1), F_i(Y_i))$ and count $V_i \in [(1-\tau)/2, (1+\tau)/2]$. This is the probability integral transform evaluated inside the atom that contains the observation, and it is unbiased for the nominal level of a continuous interval on the same distribution.

The two agree only when the pmf is smooth relative to the grid. For a RUL distribution censored at K — where 17 %

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of the C-MAPSS engines sit exactly at 125 — the endpoint rule adds the atom and part of the neighbouring bin to every interval whose endpoint lands there. Writing b_i for that surplus mass,

$$\widehat{\mathrm{PICP}}_{\mathrm{int}} - \widehat{\mathrm{PICP}}_{\mathrm{PIT}} = \frac{1}{n}\sum_i b_i \;\geq\; 0, \tag{1}$$

so the bias is one-sided and its size is set by how concentrated the pmf is near the endpoints. Table IV reports it: 0.011–0.044 at 0.90 and 0.074–0.120 at 0.50. Both numbers describe the same object, and a paper that reports one without saying which is not reporting a reproducible quantity.

The bias also runs through comparisons of *intervals*. Two intervals that contain the same pmf mass can have different endpoint coverage, because a narrower interval that avoids the cap is less likely to straddle an observation at the cap. Table I therefore reports the endpoint rule for every construction; a mass-based column would be equal across the central and shortest rows by construction and could not rank them.

B. What n engines can detect

A coverage claim is a hypothesis test. For a bin holding n engines whose true coverage is $p_0 = \tau$, reject when the exact two-sided binomial p-value of the observed count is below α; the power at an alternative p_1 is the binomial mass of that rejection region. The smallest $p_1 < \tau$ detected with power $1-\beta$ is $\delta^\star(n) = \tau - p_1$.

The binomial power function is a sawtooth in p_1, so "the largest alternative that reaches the target power" can land on an isolated spike that is not approached from below. We therefore report the deviation detected *reliably*, $\tau - \max\{p_1 : \text{power}(q) \geq 1-\beta \text{ for every } q \leq p_1\}$, which is the larger and conservative of the two definitions. It depends only on n, τ, α and β: not on the predictor, the backbone or the interval. It is a property of the *benchmark*, and it bounds what any method comparison on that benchmark can conclude.

Table V evaluates it over the five RUL strata of the test sets, with a Bonferroni correction over the five bins. Three caveats belong with the number. It assumes one query per engine and independence across engines; both hold here, because each test engine contributes its last observable window and the engines are distinct units. It is computed at $\tau = 0.90$; at 0.50 the same n detects a larger deviation, so the 50 % level is harder, not easier. And it is a statement about a test at a stated size and power, not a claim that the benchmark cannot show any difference: a paired comparison on the same engines resolves less than an unpaired one, which is why Section V-C reports the paired version.

C. Paired comparisons

Two intervals scored on the same engines are paired, and the engine is the independent unit. Replicating the predictor over seeds does not add independent engines: three seeds predicting the same 93 engines give three correlated readings of one unit. A paired test must therefore average within the engine and then test across engines, not count (engine, seed) pairs. Section V-C reports both, because the difference between them is itself a measurement result.

III. The Calibrated Object

Split conformal [12] with m calibration scores guarantees coverage for exchangeable data. CV+ [13] extends the guarantee to the case where the predictor is trained on the same data that calibrates it: for each calibration engine i held out by fold $k(i)$, the endpoint set is $\{\hat{y}_{-k(i)}(x) + R_i\}$ and the interval is a quantile of that set.

The construction calibrates the object $\hat{y}_{-k}$, a model trained on a $(1-1/K)$ fraction of the data. If what is deployed is the average of the K fold models — the natural choice, and the one used throughout — then the calibrated object and the deployed object differ. Whether that difference matters, and how much, is an empirical question; Section V-B answers it by holding everything else fixed.

A. Guarantee and scope

Proposition 1 (Asymmetric split conformal; [12], [15])**.** *Let $S_1, \ldots, S_m, S$ be exchangeable scores with $s_{(1)} \leq \cdots \leq s_{(m)}$ the ordered calibration values and the conventions $s_{(0)} = -\infty$, $s_{(m+1)} = +\infty$. For $\gamma \in [0, \alpha]$ set*

$$L_\gamma = \hat{y} + s_{(\lfloor (m+1)\gamma \rfloor)}, \qquad U_\gamma = \hat{y} + s_{(\lceil (m+1)(1-\alpha+\gamma) \rceil)}. \tag{2}$$

(i) $\Pr[Y \in [L_\gamma, U_\gamma]] \geq 1-\alpha$ *for every* $\gamma \in [0, \alpha]$.

(ii) *If $\hat{\gamma}$ is a function of a sample independent of $S_1, \ldots, S_m, S$, then (i) holds with $\gamma = \hat{\gamma}$.*

(iii) *If $\hat{\gamma}$ is chosen from $S_1, \ldots, S_m$ together with the same order statistics that (i) relies on, (i) need not hold.*

(iv) *With a continuous score density f, $\gamma \mapsto U_\gamma - L_\gamma$ is stationary where $f(L_\gamma) = f(U_\gamma)$. The equal-tailed choice $\gamma = \alpha/2$ satisfies this condition if and only if $f(q_{\alpha/2}) = f(q_{1-\alpha/2})$, which symmetry implies and skewness neither implies nor excludes. For a score law with an atom at the upper support the condition can hold on an interval of γ, in which case the width is flat rather than improved.*

Part (i) is the standard split-conformal rank argument with unequal tails; (ii) is (i) conditioned on $\hat{\gamma}$; (iii) is why a data-chosen allocation needs its own accounting, and Section V-A prices it. Part (iv) is a first-order condition, not a theorem about asymmetry: equal tails is *not* suboptimal for skewed laws in general, and a cap does not by itself make it suboptimal. Whether the condition fails, and how much width that costs, is measured in Section V-A.

Two remarks on the implementation. The grid for γ is restricted to $[1/(m+1), \alpha - 1/(m+1)]$ so that both endpoints are finite order statistics; at the two ends (2) uses $\pm\infty$ and the proof no longer describes the code. And clipping the interval to $[0, 125]$ cannot reduce coverage, because Y lies in the support by construction, so the clip is applied everywhere without further comment.

IV. Protocol

A. Data

C-MAPSS [18]: four fleets with 100, 260, 100 and 248 training engines and 93, 235, 97 and 228 test engines, 653 in total. Each test engine contributes one query, its last observable

50-cycle window, which is the query a maintenance system would issue. FD002 and FD004 mix six operating conditions.

N-CMAPSS [20]: nine fleets (DS01–DS08c, DS08d excluded on data-integrity grounds), 60 development units and 39 test units, 2 587 cycle-level test queries. The RUL cap is inactive, so the censoring convention that dominates C-MAPSS is absent. This makes it a contrast rather than a replication.

B. One deployed object, one calibration sample

The protocol is fixed in one place and every table imports it, because an earlier version of this study used three objects for the same claim and the discrepancy was not visible in the tables.

Deployed object. The mean of all 15 models (5 engine-level folds $\times$ 3 seeds). One object, not an average over three seed-specific objects.

Calibration object. One score per *engine*. For the matched arm $R_i = y_i - \frac{1}{S}\sum_s \hat{y}_{-k(i),s}(x_i)$, the average of the three seed models that never saw engine i — the largest clean ensemble the fold structure can produce, against the 15 that are deployed. For the CV+ arm $R_i = y_i - \hat{y}_{-k(i),s(i)}(x_i)$, one model per engine, with the seed cycling so all three contribute.

Scale. $\sigma(\hat{y}) = \mathrm{softplus}(a + b\hat{y}/125) \cdot 20 + 1$, fitted by Gaussian negative log-likelihood on that arm's own residuals, so no arm borrows another's scale.

Tail. $\gamma = \alpha/2$ unless stated. Where γ is chosen from data it is chosen on an interior grid and, in the *split* variant, on a fixed half of the calibration engines, with the interval built on the other half.

Metrics. Coverage is the integer-endpoint frequency for every construction; the mass-based figure is carried as a diagnostic. Width is the mean of $U - L$ in cycles. The interval score of Winkler [21], a standard prognostic metric [19], is reported at $\alpha = 0.10$ because it penalises width and misses jointly. Uncertainty on a coverage is the Wilson interval [22]; uncertainty on a width, and on a difference of widths, is a bootstrap over *engines* (2 000 resamples).

No test label is read during calibration. On N-CMAPSS this required a change: an earlier version calibrated the pooled construction on other fleets' *test* residuals, which is a leak. The pooled result here is fixed on all development residuals ($n = 3\,995$ cycles) and applied once to the test queries.

C. Simulation

The benchmark cannot check nominal coverage, so a simulation with known truth does. The generator draws $Y_0 = 125(1-u)$, $u \sim \mathrm{U}(0,1)$, a lattice observation $Y = \min\{\mathrm{round}(Y_0 + \varepsilon), 125\}$ with $\varepsilon \sim N(0, 9^2)$, and a deliberately biased predictor $\hat{y} = 1.06Y_0 + \eta + 4$ with $\eta \sim N(0, 5.5^2)$, left unclipped so that over-prediction near the cap produces the same one-sided residuals as C-MAPSS. Each configuration is repeated 200 times with a fresh test set of 4 000 engines.

V. Results

Table I is the whole comparison on one protocol. Three readings are available and only one of them is favourable to the conformal family.

Sharpness. The frozen 66-parameter layer is the narrowest object at 30.9 cycles and has the best interval score at 41.0. Its shortest-interval variant is narrower still at 27.7, but that is a different object (Section V-A). Among the conformal constructions the spread is 31.4–36.0 cycles, so the layer remains the sharpest thing in the table. This is not the result the study was designed to find and it is the one that constrains the conclusions most.

Calibration stability. The layer's $|\Delta|$ is 0.025; the conformal family spans 0.016–0.030. An earlier draft reported a factor-of-three advantage for the conformal family; it does not survive the switch to a single scoring rule, on which the layer over-covers on three fleets and under-covers on one, and its spread is the same size as the others'.

Cost. The matched free-tail construction at 31.9 cycles is 3.2 % wider than the layer's 30.9 and 2.3 Winkler points worse. That is the honest size of the gap: small, and in the layer's favour.

A. The tail allocation, and what a guarantee costs

Between equal tails and a free allocation on the matched arm the widths are 32.2 and 31.9 (+0.9 % on average), with per-fleet changes of $+5.0$, $+4.2$, 0.0 and -4.3 %: the sign reverses on FD004 and the mean hides it. Proposition 1(iv) explains why this is not a paradox — $f(L) = f(U)$ is an equality that a given score law may or may not satisfy — and the simulation shows what happens when it is satisfied.

There, where calibration and deployment are exchangeable and the score law is known, the free tail saves 16 % of width at nominal 0.90 (31.3 against 37.4 cycles at coverage 0.901 for both) and 22 % at 0.50 (13.1 against 16.6, at 0.506 and 0.510). The mechanism is real. On the benchmark it does not materialise, and what differs between the two settings is the exchangeability assumption, not the construction.

The guarantee is a separate matter from the width. Choosing γ on the same sample that supplies the order statistics is a selection, and Proposition 1(iii) says the bound does not carry over. The *half-split* variant restores it honestly: γ is chosen on a fixed half of the calibration engines and the interval is built on the other half, so γ is independent of the scores it is applied to and (ii) applies. The price is 13 % of width (31.9 to 36.0) and 2.3 Winkler points. A paper that reports a free tail without that split is reporting the width of a selected interval against the guarantee of a fixed one.

B. The calibrated object, controlled

Table II is the experiment this paper was built around. Everything that could confound the comparison is fixed; the calibration residuals are the only thing that varies. The matched object is narrower on all four fleets by 2.7–10.8 %, mean 5.4 %. It costs coverage on FD003 (0.907 to 0.887) and buys it on FD002 and FD004.

TABLE I

CONSTRUCTIONS ON THE FOUR C-MAPSS FLEETS AT NOMINAL 0.90, ONE PROTOCOL, ONE DEPLOYED OBJECT (15 MODELS). COVERAGE IS THE INTEGER-ENDPOINT FREQUENCY ON 653 TEST ENGINES; WIDTH IS IN CYCLES; $|\Delta|$ IS THE MEAN ABSOLUTE DEVIATION FROM 0.90 ACROSS FLEETS. WINKLER IS THE INTERVAL SCORE AT $\alpha = 0.10$, LOWER IS BETTER. THE MASS-BASED COVERAGE OF THE TWO LAYER ROWS IS 0.90 FOR BOTH BY CONSTRUCTION (0.925/0.869/0.932/0.842 AS A RANDOMISED-PIT DIAGNOSTIC), WHICH IS WHY IT CANNOT RANK THEM.

Construction	FD001	FD002	FD003	FD004	$\lvert\Delta\rvert$	Winkler	width
Layer, central	0.935	0.902	0.948	0.886	0.025	**41.0**	30.9
Layer, shortest	0.914	0.885	0.907	0.873	0.016	43.7	**27.7**
CV+, symmetric	0.892	0.936	0.897	0.934	0.020	44.9	31.9
CV+, equal tails	0.892	0.936	0.887	0.925	0.021	46.5	32.7
Matched, symmetric	0.882	0.928	0.897	0.947	0.024	45.3	31.4
Matched, equal tails	0.892	0.923	0.887	0.930	**0.019**	46.8	32.2
Matched, free tail	0.914	0.919	0.887	0.917	0.016	45.1	31.9
Matched, free tail, half-split	0.925	0.923	0.876	0.947	0.030	47.4	36.0
Matched, free tail, no scale	0.892	0.940	0.907	0.895	0.015	49.6	34.6

TABLE II

CONTROLLED OBJECT EXPERIMENT. DEPLOYMENT CENTRE, SCORE SCALE, TAIL ALLOCATION ($\gamma = \alpha/2$), TEST ENGINES AND CALIBRATION ENGINES ARE IDENTICAL ACROSS THE TWO BLOCKS; ONLY THE RESIDUALS THAT ENTER THE SCORE SAMPLE CHANGE. COVERAGE IS THE INTEGER-ENDPOINT FREQUENCY AT NOMINAL 0.90; WIDTH IN CYCLES. THE LAST LINE GIVES THE RELATIVE WIDTH CHANGE, POSITIVE MEANING THE MATCHED OBJECT IS NARROWER. DEPLOYMENT IS THE 15-MODEL MEAN IN BOTH BLOCKS AND THE CALIBRATION SIZE IS ONE SCORE PER ENGINE IN BOTH, SO THE DIFFERENCE IS THE OBJECT AND NOT n; REPLICATING THE MATCHED SCORE THREE TIMES CHANGES NO WIDTH TO ONE DECIMAL.

Calibration object		FD001	FD002	FD003	FD004
one clean model (CV+)	coverage	0.892	0.915	0.907	0.925
	width	30.9	34.7	28.8	41.3
three clean models, averaged	coverage	0.892	0.923	0.887	0.930
	width	29.5	33.4	25.7	40.2
width change from matching		-4.5	-3.7	-10.8	-2.7

The number is smaller than the 11 % an earlier draft reported, and the difference is accounted for. That draft compared CV+ with a symmetric band against a matched construction with a free tail *and* variance normalisation, so its 11 % mixed three effects. On this protocol the object is worth 5.4 %, the free tail about 1 %, and the normalisation 0.0–0.2 Winkler points once the object is matched. The components are separately visible in Table I.

Two further controls, not tabulated. Repeating the comparison with the score replicated three times, so both arms have the same calibration size, changes no width to one decimal: the effect is the object, not n. And deploying the five-model average of the earlier protocol instead of the 15-model mean changes the widths by less than 0.2 cycles: the centre was never the confound.

What does not replicate is the sign. Across the four nominal levels the matched object is narrower by 2.4, 5.6, 2.3 and 6.1 % on average, but at every level at least one fleet reverses (FD002 at 0.90, FD001 at 0.80, FD003 at 0.95). On the second backbone the four per-fleet changes are $+3.7$, -7.5, -0.6 and $+0.2$ %. A 2–6 % average effect with a fleet-dependent sign is a real effect that cannot support a ranking, and it is why this paper reports the experiment rather than a favourite construction.

C. Paired test, clustered by engine

Table III compares the two tail rules on identical scores. No fleet rejects at 5 % under either test, and 4–8 engines per fleet change status. The two rules are therefore indistinguishable on this benchmark, which is consistent with the 1 % width difference in Table I and inconsistent with an earlier draft's claim of significance in both directions. That claim came from counting three seeds as three independent observations; the naive column here is already non-significant, and the clustered one is wider still.

TABLE III

FREE TAIL AGAINST SYMMETRIC BAND ON THE MATCHED ARM, SAME SCORES AND SAME NOMINAL LEVEL. THE NAIVE TEST COUNTS (ENGINE, SEED) PAIRS; THE CLUSTERED VERSION AVERAGES OVER SEEDS INSIDE AN ENGINE AND TESTS ACROSS ENGINES (10 000 SIGN FLIPS, 2 000 BOOTSTRAP RESAMPLES OVER ENGINES).

Fleet	naive p	pairs	clustered p	difference [95 % CI]	engines
FD001	0.267	279	0.121	$+0.018$ [0.000, 0.036]	7/93
FD002	1.000	705	1.000	-0.000 [-0.010, 0.011]	8/235
FD003	0.219	291	0.308	-0.014 [-0.034, 0.003]	5/97
FD004	0.125	684	0.127	$+0.006$ [0.001, 0.012]	4/228

The value of the table is methodological. A paired design can resolve smaller differences than the unpaired bound of Table V, but only if the discordance is counted on the independent unit. On 93 engines a difference carried by 7 engines is not evidence, and seed replication does not change that.

D. The independent benchmark, without the leak

On N-CMAPSS, with the rule fixed on development residuals and applied once, the free-tail construction returns 0.934

TABLE IV
COVERAGE BIAS OF THE INTEGER-ENDPOINT RULE ON THE FROZEN LAYER. SAME PREDICTIVE DISTRIBUTION, TWO SCORING CONVENTIONS.

	at nominal 0.90		at nominal 0.50	
Fleet	PIT	integer	PIT	integer
FD001	0.925	0.935	0.517	0.591
FD002	0.869	0.902	0.523	0.643
FD003	0.932	0.948	0.524	0.608
FD004	0.842	0.886	0.420	0.535
bias	$+0.011$ to $+0.044$		$+0.074$ to $+0.120$	

(95 % unit-bootstrap interval $[0.895, 0.968]$) at width 47.1 and the symmetric band 0.947 at width 47.4: a 1 % saving, not the 19 % reported earlier. At nominal 0.50 the free tail is 7 % *wider* (14.6 against 13.6). Fleet-balanced and query-weighted figures agree to within 0.3 %.

The earlier number came from calibrating on other fleets' test residuals. Development residuals are systematically larger than test residuals on this benchmark (pooled dev RMSE 14.7 against roughly 12 on test), so a calibration sample drawn from the test population is narrower and exaggerates the asymmetry of the score law. Both arms inherit the narrower sample; the asymmetric arm benefits more because it can place its endpoints where the skew is. Removing the leak removes the mechanism with it.

What survives is a smaller observation: dev-calibrated intervals over-cover on test (0.934 and 0.947 against a nominal 0.90), because development predictions are worse than deployed ones. That is the same object mismatch as Section V-B, appearing as a level shift rather than a width change, on a benchmark where the calibration and deployment populations genuinely differ.

E. The second backbone

Every number above comes from one network. Recomputing the identical protocol on a two-layer bidirectional LSTM with additive attention pooling (119 490 parameters against 41 858, dev RMSE 13.1/15.3/12.7/15.4 against 11.1/13.9/10.5/13.9) reproduces the layer's advantage — the layer is 20–30 % narrower than any conformal construction on three fleets — but not the object effect: the four per-fleet changes are $+3.7$, -7.5, -0.6 and $+0.2$ %, so the sign is set by the fleet rather than by the backbone. Coverage under the second backbone is 0.753–0.953, a spread twice that of the primary.

That is the controlled demonstration of the paper's thesis in its weakest form: change the predictor and the coverage numbers move by 0.13 with the rules held fixed, while the width ordering among constructions is preserved. Coverage at this scale measures the predictor–population pair; width measures the construction.

VI. DISCUSSION

A. What the measurements say about the constructions

The conformal family does not beat the parametric layer here, and the paper reports that rather than selecting a fleet on which it does. Three things follow.

TABLE V
WHAT A BIN CAN PROVE. BIN SIZES ARE THE FIVE RUL STRATA OF THE TEST SETS; $\delta^\star$ IS THE RELIABLY DETECTABLE UNDER-COVERAGE AT FAMILY-WISE $\alpha = 0.05$ OVER FIVE BINS, 80 % POWER, ONE QUERY PER ENGINE.

	engines per bin				
Fleet	$[0, 25)$	$[25, 50)$	$[50, 75)$	$[75, 100)$	$[100, 125]$
FD001	19	11	13	22	28
FD002	58	29	31	45	72
FD003	15	14	20	18	30
FD004	49	30	26	41	82
pooled	141	84	90	126	212

$\delta^\star$ at $n = 11$: 0.44 $n = 82$: 0.14 $n = 141$: 0.10 $n = 212$: 0.08
engines needed: 143 for $\delta^\star = 0.10$, 238 for 0.075, 495 for 0.05

A layer fitted to the residual law has an advantage conformal methods cannot borrow: it is heteroscedastic by construction, so its width follows the prediction rather than being fixed by a single quantile. The variance normalisation in Table I is an attempt to borrow it, and it works — removing it costs 2.8–5.0 Winkler points — but it recovers only part of the layer's 41.0 against 45.1.

The object effect is real at 2–6 % and is the one place where the conformal literature's standard construction is measurably wrong for this deployment pattern. It is not worth 11 % and its sign is not stable across fleets. A paper that claims otherwise on four fleets is claiming a mean of a bimodal quantity.

The free tail is a convention whose value depends on the score law, not a free lunch. In simulation it is worth 16–22 %; on the benchmark 0–5 % and reversing; and the price of making it legitimate is 13 % of width. Those three numbers should be reported together or not at all.

B. What should be reported

Four rules, each a measurement fix rather than a modelling choice. (i) Report coverage with its interval and the number of engines, not only the rate. (ii) State the scoring rule: integer endpoints and randomised PIT differ by 0.011–0.044 at 0.90 on the same distribution, so a table that does not say which is comparing an unknown to an unknown. (iii) Report width, and report it per fleet, because the fleet is a larger source of variation than the method ($|\Delta|$ 0.09 across fleets for the layer on the PIT rule against 0.02 between constructions). (iv) State which object was calibrated: if the deployed predictor is an ensemble and the calibration residuals come from one member, the interval is calibrated for a predictor that is not in service, and the difference is 2–11 % of width here with a fleet-dependent sign.

C. Limitations

Both datasets are run-to-failure simulators; there is no real sensor data in this study. The public run-to-failure bearing datasets cannot support a coverage claim at 143 engines per bin — XJTU-SY has 15 units, PRONOSTIA 17 — so the natural next experiment is not available at the sample size the measurement requires, which is itself the paper's point.

The 50 % level is not resolved by any construction here: every arm lands between 0.34 and 0.66 with $|\Delta|$ of 0.03–0.09, and the ordering changes. We report the failure rather than selecting the seed that hides it.

The detectability bound is stated for one test at a stated size and power with one query per engine. It is not a statement that no comparison on this benchmark can be informative; the paired design of Section V-C resolves less than the unpaired bound, and a different binning would give a different number.

VII. Conclusion

RUL intervals are reported as a coverage and a width, and this paper has measured how much of each is convention. On 653 C-MAPSS engines and 2 587 N-CMAPSS queries: the integer-endpoint rule inflates coverage by 0.011–0.044 at 0.90 and 0.074–0.120 at 0.50 with no dependence on the model; an RUL bin of 11–82 engines detects only 0.14–0.44 and 143 per bin are needed for 0.10, more than any of these fleets provides; calibrating a fold model while deploying an ensemble costs 2–11 % of width depending on the fleet, with the sign reversing on one; and a free tail saves 16–22 % in simulation but 0–5 % here, with a 13 % price if its allocation is made independent of the scores it is applied to.

The parametric layer is sharper than any conformal construction in the table (30.9 against 31.4–36.0 cycles) and scores better on the interval score (41.0 against 44.9–49.6). We report that because it bounds the claim: what this study establishes is a measurement framework in which such comparisons can be made reproducibly, not a construction that beats the state of the art. The broader conclusion is that the field's uncertainty numbers are limited by reporting practice more than by modelling, and that the practices in question are cheap to fix.

Appendix A
Proof of Proposition 1

Assume continuous scores, so ties have probability zero; with ties, randomise the rank. Let $R = \#\{i \le m : S_i \le S\}$. Under exchangeability of $(S_1, \dots, S_m, S)$ the rank of S among the $m+1$ values is uniform, so $\Pr[R = r] = 1/(m+1)$ for $r = 0, \dots, m$.

(i) $S < L_\gamma$ holds iff fewer than $\lfloor (m+1)\gamma \rfloor$ calibration scores lie below S, i.e. iff $R \le \lfloor (m+1)\gamma \rfloor - 1$, hence $\Pr[S < L_\gamma] \le \gamma$. Likewise $S > U_\gamma$ iff $R \ge \lceil (m+1)(1-\alpha+\gamma) \rceil$, hence $\Pr[S > U_\gamma] \le \big(m+1-\lceil (m+1)(1-\alpha+\gamma) \rceil\big)/(m+1) \le \alpha - \gamma$. The two events are disjoint, so the miss probability is at most α. This is the argument of [12] with unequal tails, as used by [15]; it is restated to fix the scope, not claimed as new.

For (2) to involve finite order statistics at both ends one needs $\lfloor (m+1)\gamma \rfloor \ge 1$ and $\lceil (m+1)(1-\alpha+\gamma) \rceil \le m$, i.e. $\gamma \in [1/(m+1), \alpha - 1/(m+1)]$; outside that interval the endpoints are $\pm\infty$ and the implementation must say so.

(ii) Condition on $\hat{\gamma}$, which by assumption is independent of the scores. Applying (i) at the realised value gives the bound.

(iii) Let $\hat{\gamma}$ minimise the empirical width. The event $\{\hat{\gamma} = \gamma_0\}$ is determined by the same order statistics that (i) integrates over, so the conditional coverage at γ_0 is not the unconditional probability that (i) bounds. A union bound over a grid of G values would cost an additional slack per candidate; the practical remedies are the half-split of Section V-A or a sample-splitting penalty.

(iv) With a continuous score density, $U_\gamma - L_\gamma \to q_{1-\alpha+\gamma} - q_\gamma$ and

$$\frac{d}{d\gamma}\big(q_{1-\alpha+\gamma} - q_\gamma\big) = \frac{1}{f(q_{1-\alpha+\gamma})} - \frac{1}{f(q_\gamma)},$$

which vanishes iff the two densities are equal. At $\gamma = \alpha/2$ this reads $f(q_{\alpha/2}) = f(q_{1-\alpha/2})$. Symmetry of f about the median is sufficient; skewness is neither necessary nor sufficient for the condition to fail, since a skewed density may still take equal values at the two quantiles. If f has an atom at the upper support, the stationarity condition can hold on an interval of γ and the width is flat there rather than minimised at a point.

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